

Linux System Rescue Guide

Root Password Reset & GRUB Repair via Live USB

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Introduction

This guide details two advanced recovery techniques for Linux systems using a **Live USB**. Both methods rely on the same fundamental principle: booting a temporary operating system to access and modify the files on the computer's hard drive while the main system is "asleep."

Warning: These methods grant total control over the system. Ensure you have identified the correct drive partitions before executing write commands.

Glossary of Terms

Before proceeding, it is crucial to understand the tools and concepts used in these procedures.

Live USB A USB flash drive containing a full, bootable Linux operating system (like Ubuntu, Fedora, or Mint). It runs entirely in the computer's RAM and does not modify the hard drive unless you explicitly tell it to.

Partition A defined section of the hard drive. Your Linux installation typically lives on a specific partition (e.g., `/dev/sda2`).

Mounting The act of attaching a storage device (like your hard drive partition) to a specific folder in the current file system so its contents can be accessed.

Chroot (Change Root) A command that changes the apparent root directory for the current running process. It effectively "traps" your terminal inside the mounted hard drive, making the shell behave as if the hard drive's system were the one currently running.

GRUB The **GRand Unified Bootloader**. This is the first program that runs when the computer starts; it loads the Linux Kernel. If GRUB is corrupted, the system cannot boot.

Bind Mount A special type of mounting that mirrors a directory. We use it to "loan" the Live USB's hardware connections (like `/dev` and `/proc`) to the broken system so it can function during the repair.

Method 1: Resetting a Forgotten Root Password

Use this method if you have lost your password and cannot log in.

Step 1: Preparation

1. Insert your Linux Live USB and boot the computer from it.
2. Select **"Try Linux"** (do not install).
3. Open a terminal window.
4. Switch to the root user in the Live session:

```
sudo -i
```

Step 2: Identification

List all storage devices to find your main Linux partition. Look for the largest partition.

```
lsblk
```

*Assumption for this guide: Your main system is on **/dev/sda2**.*

Step 3: Accessing the Drive

Mount your hard drive partition to the `/mnt` directory.

```
mount /dev/sda2 /mnt
```

Step 4: Entering the System (Chroot)

Move your terminal session **"inside" the hard drive**.

```
chroot /mnt
```

Your command prompt should change. You are now effectively "logged in" to the hard drive as root.

Step 5: The Fix

Change the password. You do not need to know the old one.

```
passwd root
```

Type the new password when prompted. If you need to reset a specific user's password (e.g., "john"), type `passwd john`.

Step 6: Exit

1. Type **exit** to leave the chroot environment.
2. Unmount the drive: `umount /mnt`
3. Reboot the computer and remove the USB.

Method 2: Reinstalling GRUB (Boot Repair)

Use this method if your computer boots to a black screen, a "GRUB Rescue" prompt, or directly into Windows (ignoring Linux).

Step 1: Preparation & Identification

Follow Steps 1 and 2 from the Password Reset method to boot the Live USB and identify your partition (`/dev/sda2`).

Step 2: Mounting the System

Mount your root partition:

```
mount /dev/sda2 /mnt
```

CRITICAL FOR UEFI SYSTEMS: If you have a modern computer (UEFI), you must also mount your EFI partition. This is usually a small partition (approx 100MB-500MB) at the start of the disk (e.g., `/dev/sda1`).

```
mount /dev/sda1 /mnt/boot/efi
```

Step 3: Binding Hardware (The "Life Support")

The broken system needs access to the hardware information to install bootloaders. We "bind" the Live USB's system folders to the mounted drive. Run this exact command:

```
for i in /dev /dev/pts /proc /sys /run; do sudo mount -B $i /mnt$i; done
```

Step 4: Enter the System (Chroot)

```
chroot /mnt
```

Step 5: Reinstall Bootloader

Select the command based on your system type:

Option A: For Legacy/MBR Systems (Older computers) Install GRUB to the **drive** (`sda`), not the partition (`sda1`).

```
grub-install /dev/sda
```

Option B: For UEFI/GPT Systems (Modern computers)

```
grub-install
```

Step 6: Update Configuration

Generate the new menu configuration file.

```
update-grub
```

Note: On non-Debian systems (like Fedora/Arch), use:

```
grub-mkconfig -o /boot/grub/grub.cfg
```

Step 7: Exit

1. Type `exit` to leave the chroot.
2. Unmount everything: `umount -R /mnt`
3. Reboot.